MAT 3007 – Optimization

Convex Optimization

Lecture 21

Instructor: Junchi Yang

CUHK-Shenzhen

Recap

Convex Functions

Definition: Convex Function

A function f on a convex set Ω is said to be convex if for every $\boldsymbol{x}, \boldsymbol{y} \in \Omega$ and any $0 \leq \lambda \leq 1$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

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- If f be twice cont. differentiable and $\Omega \subseteq \mathbb{R}^n$ is convex and open, then f is convex on Ω if and only if its Hessian matrix $\nabla^2 f(\boldsymbol{x})$ is positive semidefinite for all $\boldsymbol{x} \in \Omega$.

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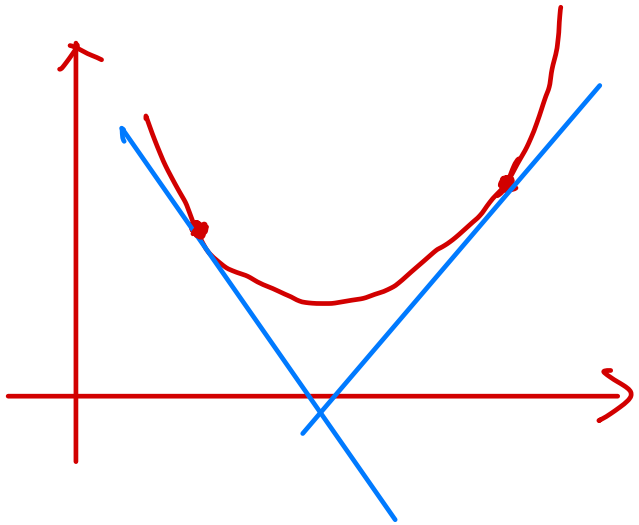
- If f be twice cont. differentiable and $\Omega \subseteq \mathbb{R}^n$ is convex and open, then f is convex on Ω if and only if its Hessian matrix $\nabla^2 f(\boldsymbol{x})$ is positive semidefinite for all $\boldsymbol{x} \in \Omega$.
- We call f a concave function if and only if $-f$ is convex.

An Equivalent Definition for Convex Function

Theorem: First-Order Characterization of Convexity

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable. f is convex ($\mathbb{R}^n$) if and only if

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$$



Geometrical Interpretation:
graph is above the tangent line.

An Equivalent Definition for Convex Function

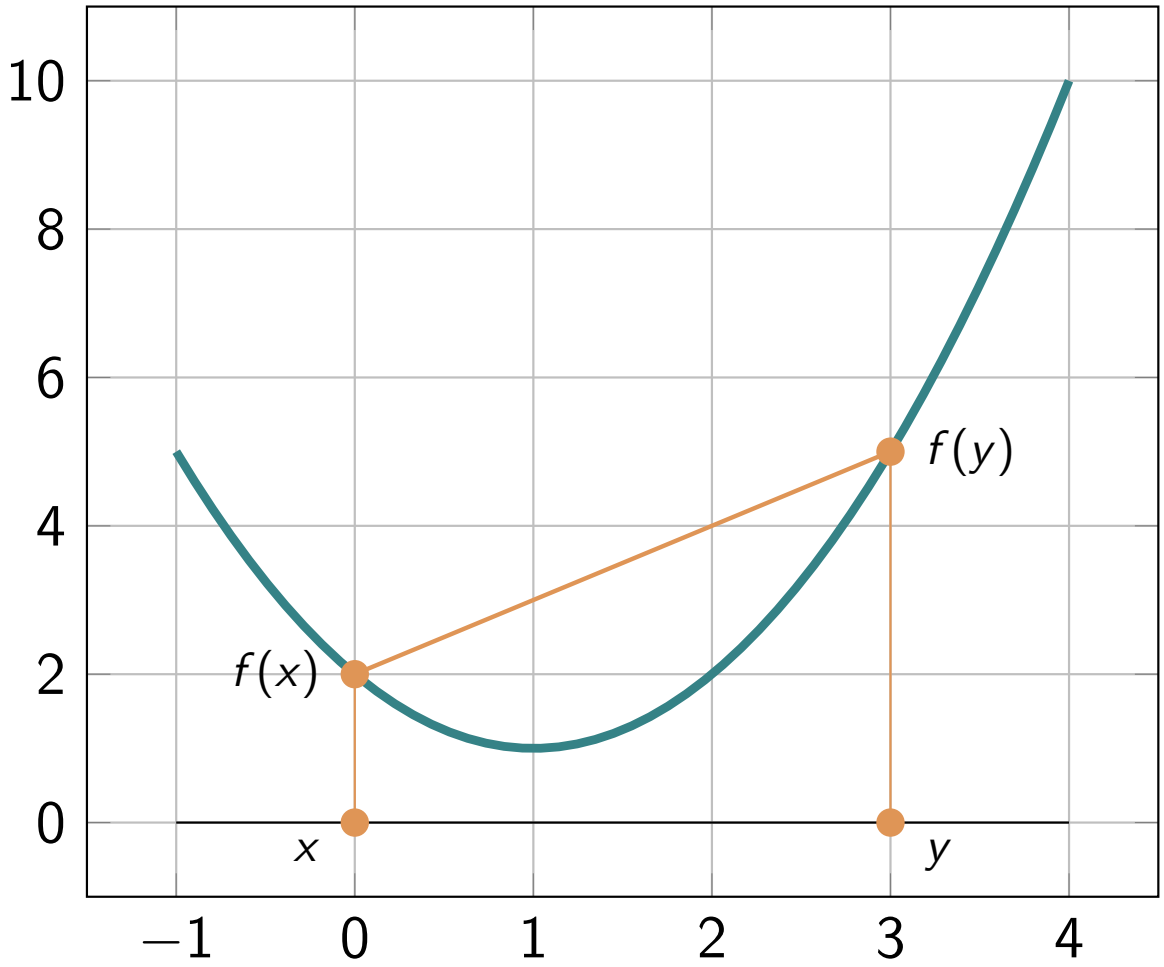
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- This is an equivalent definition for convex function when f is differentiable. This first-order characterization is often used for analysis.
- See section 3.1.3 of "Boyd, S., Boyd, S. P., & Vandenberghe, L. (2004). Convex optimization. Cambridge university press" for a proof to the above theorem.

Illustration of Convex Functions



Sum Rule:

- If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex (concave) functions, then $a_1 f_1 + \cdots + a_m f_m$ is a convex (concave) function.

Sum Rule:

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Composition:

- If $f : \mathbb{R}^m \to \mathbb{R}$ is convex (concave) and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{b} \in \mathbb{R}^m$ are given, then $g : \mathbb{R}^n \to \mathbb{R}$, $g(\boldsymbol{x}) := f(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$, is convex (concave).

Why $g(x) = f(Ax+b)$ is convex ?

proof: For any two points x and y, and any $\lambda \in [0,1]$

$$g(\lambda x + (1-\lambda)y) = f(A(\lambda x + (1-\lambda)y) + b)$$

$$= f(\lambda Ax + (1-\lambda)Ay + b)$$

$$= f(\lambda(Ax+b) + (1-\lambda)(Ay+b))$$

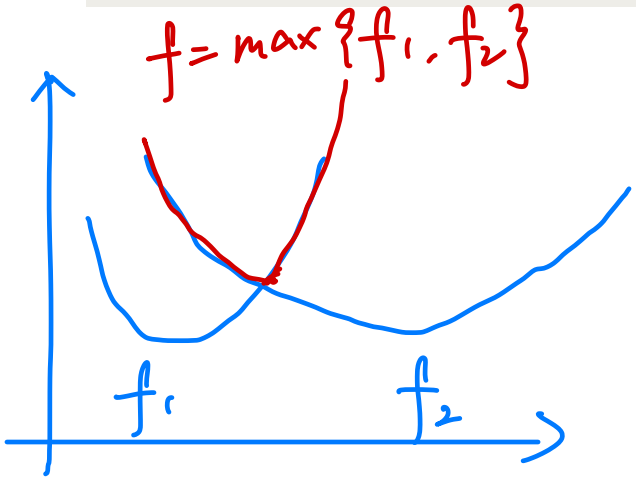
$$\overset{f \text{ is convex}}{\leq} \lambda f(Ax+b) + (1-\lambda) f(Ay+b)$$

$$= \lambda g(x) + (1-\lambda) g(y)$$

□

Lemma: Taking Maximum

If $f_1, \ldots, f_m$ are convex functions, then
$f(\boldsymbol{x}) = \max\{f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})\}$ is a convex function.



proof: for each i, since f_i is convex

$$f_i(\lambda x + (1-\lambda)y) \le \lambda f_i(x) + (1-\lambda) f_i(y)$$

take max over all i

$$\max_i f_i(\lambda x + (1-\lambda)y) \le \max_i \left[\lambda f_i(x) + (1-\lambda) f_i(y)\right]$$

$$= f(\lambda x + (1-\lambda)y) \qquad \le \lambda \max_i f_i(x) + (1-\lambda) \max_i f_i(y)$$

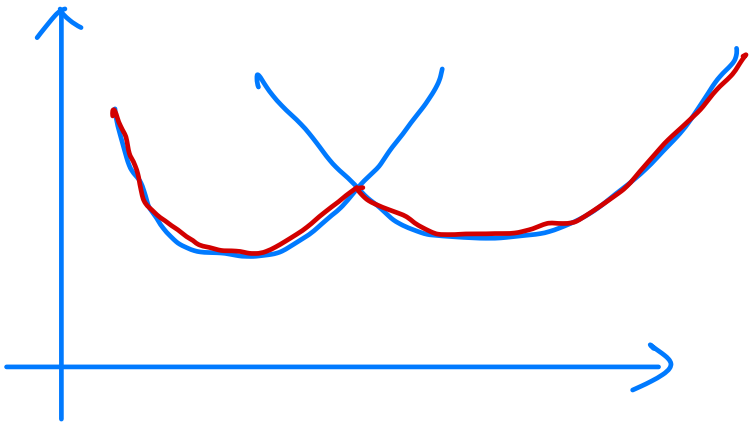
$$= \lambda f(x) + (1-\lambda) f(y)$$

Further Properties: Max & Min

Lemma: Taking Maximum

If $f_1, \ldots, f_m$ are convex functions, then
$f(\boldsymbol{x}) = \max\{f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})\}$ is a convex function.

- **Examples:** $|x| = \max\{-x, x\}$, $\max\{\boldsymbol{a}_i^\top \boldsymbol{x} + b_i\}$.
- This can be extended to $\sup_{i \in \mathcal{I}} f_i$ where $\{f_i\}_{i \in \mathcal{I}}$ is a family of convex functions.



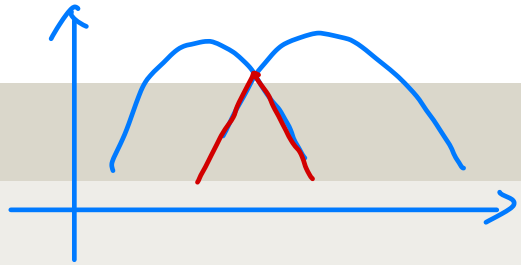
Note: $\min_i \{f_i(x)\}$ may not be convex.

Further Properties: Max & Min

Lemma: Taking Maximum

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Lemma: Taking Minimum

If $f_1, \ldots, f_m$ are concave function, then
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- **Examples:** $-|x| = \min\{-x, x\}$, $\min\{\boldsymbol{a}_i^\top \boldsymbol{x} + b_i\}$.

Another Example: Linear Programming

Consider the linear program

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

Given $\boldsymbol{A}$ and $\boldsymbol{b}$ fixed, the optimal value function is a function of $\boldsymbol{c}$. We denote the function by $V(\boldsymbol{c})$.

- In sensitivity analysis, we studied how $V(\boldsymbol{c})$ changes with $\boldsymbol{c}$.

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Theorem: Properties of V

V is a concave function of $\boldsymbol{c}$.

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Given $\boldsymbol{A}$ and $\boldsymbol{b}$ fixed, the optimal value function is a function of $\boldsymbol{c}$. We denote the function by $V(\boldsymbol{c})$.

- In sensitivity analysis, we studied how $V(\boldsymbol{c})$ changes with $\boldsymbol{c}$.

Theorem: Properties of V

V is a concave function of $\boldsymbol{c}$.

- V is the minimum of a set of linear functions

$$V(\boldsymbol{c}) = \min_{\{\boldsymbol{x}:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b},\boldsymbol{x}\geq\boldsymbol{0}\}} \{\boldsymbol{c}^\top \boldsymbol{x}\}.$$

Convex Optimization Problem

We call the optimization problems of the form:

- Minimize a convex function over a convex feasible region
- Maximize a concave function over a convex feasible region

convex optimization problems.

Otherwise, the problem is called a non-convex optimization problem.

- Convex problems are generally easier to solve efficiently.

Convex Constraints

Consider $\min_x f(x)$

s.t $g_i(x) \le 0$

$\lambda_i(x) = 0$

When is the feasible region convex?

Lemma: Convex Level Sets

Let g be a convex function. Then, for any c, the level set $L_{\leq c} = \{\boldsymbol{x} : \boldsymbol{g}(\boldsymbol{x}) \leq c\}$ is a convex set.

proof: If $x, y \in L_{\leq c}$, then $g(x) \leq c$ and $g(y) \leq c$

$\forall \lambda \in [0,1]$, consider $\lambda x + (1-\lambda) y$

$$g(\lambda x + (1-\lambda) y) \leq \lambda g(x) + (1-\lambda) g(y)$$

$$\leq \lambda c + (1-\lambda) c$$

$$= c$$

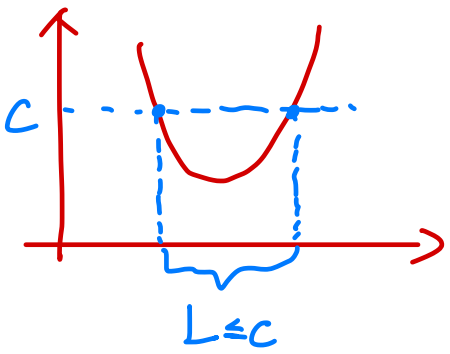
$$\Rightarrow \lambda x + (1-\lambda) y \in L_{\leq c}$$

Lemma: Convex Level Sets

Let g be a convex function. Then, for any c, the level set $L_{\leq c} = \{\boldsymbol{x} : f(\boldsymbol{x}) \leq c\}$ is a convex set.

Observation:

- If we have constraints of the form $g(\boldsymbol{x}) \leq 0$ and g is convex, then this is a convex constraint.



$\{x : g(x) \leq c\}$ is convex.

$\{x : g(x) = c\}$ may be nonconvex.

$\{x : g(x) \geq c\}$ may be nonconvex.

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Observation:

- If we have constraints of the form $g(\boldsymbol{x}) \leq 0$ and g is convex, then this is a convex constraint.
- Linear constraints are always convex constraints.

$\{x : a^T x \leq b\}$ is convex

$\{x : a^T x = b\}$ is also convex because it's equivalent to $\{x : a^T x \leq b \text{ and } -a^T x \leq -b\}$.

$\{x : a^T x \geq b\}$ is also convex.

Lemma: Convex Level Sets

Let g be a convex function. Then, for any c, the level set $L_{\leq c} = \{\boldsymbol{x} : f(\boldsymbol{x}) \leq c\}$ is a convex set.

Observation:

- If we have constraints of the form $g(\boldsymbol{x}) \leq 0$ and g is convex, then this is a convex constraint.
- Linear constraints are always convex constraints.
- Sometimes, even if a constraint does not appear to be in the above form, it still could be a convex constraint (this lemma is just sufficient but not necessary).

See an example in Slide 16.

Intersection of Convex Sets

Lemma

The intersection of finite number of convex sets, $C_1, C_2, \ldots, C_m$,

$$C = \bigcap_{i=1}^{m} C_i = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{x} \in C_i \text{ for all } i = 1, \ldots, m\}$$

is a convex set.

proof: If $x, y \in C$, then $x \in C_i$, $i = 1, \ldots, m$
$y \in C_i$, $i = 1, \ldots, m$

Because C is convex, $\lambda x + (1-\lambda) y \in C_i$
$\Rightarrow \lambda x + (1-\lambda) y \in \bigcap C_i$ □

Remark: This is also true for infinite number of sets, $C = \bigcap_{i \in I} C_i$.
A similar proof holds.

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is a convex set.

Remark:

- If we have multiple constraints of the form $g_i(\boldsymbol{x}) \leq 0$ for $i = 1, \ldots, m$, and each g_i is convex, then this is a convex constraint.

Example — I

Is this a convex optimization problem?

$$\begin{aligned} \text{minimize} \quad & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ \text{subject to} \quad & x_1^2 + x_2^2 \leq 5 \\ & 3x_1 + x_2 \geq 3 \end{aligned}$$

Example — I

Is this a convex optimization problem?

check f is convex

$$\begin{aligned} \text{minimize} \quad & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ \text{subject to} \quad & x_1^2 + x_2^2 \le 5 \\ & 3x_1 + x_2 \ge 3 \end{aligned}$$

$g_1(x) = x_1^2 + x_2^2 - 5$

$g_2(x) = -3x - x_2 + 3$

each g_i is convex

- **Answer:** Yes.

Example — I

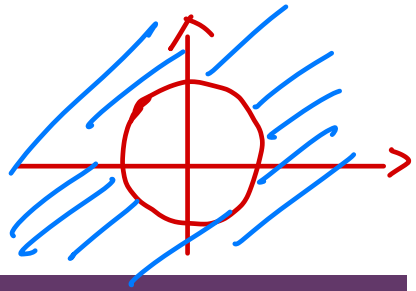
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- **Answer:** Yes.

What if we change the constraint $x_1^2 + x_2^2 \le 5$ to $x_1^2 + x_2^2 \ge 5$?

- Then it no longer is a convex optimization problem.



Example — II

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} - \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{C}\boldsymbol{x} \geq \boldsymbol{d} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

- The constraints are linear. ⟵ convex constraint.
- It is a convex optimization problem if and only if $\boldsymbol{Q}$ is PSD.

→ check the convexity of f by computing its Hessian.

Example — III

$$\begin{aligned} \text{maximize}_{x,y,z} \quad & xyz \\ \text{s.t.} \quad & x + 2y + 3z \le 3 \\ & x, y, z \ge 0 \end{aligned}$$

Note: although xyz is linear in x, y, z individually, it does not mean it's concave in them jointly.

Why $f(x,y,z) = xyz$ is not concave?

$$\nabla f(x,y,z) = \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} \quad \nabla^2 f(x,y,z) = \begin{pmatrix} 0 & z & y \\ z & 0 & x \\ y & x & 0 \end{pmatrix}$$

$$\nabla^2 f(1,1,1) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

For a vector $u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$

$u^T \nabla^2 f(1,1,1)\, u$

$= 2u_1u_2 + 2u_2u_3 + 2u_1u_3$

> 0 with $u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$\Rightarrow \nabla^2 f(1,1,1)$ is not NSD.

$\Rightarrow f$ is not concave.

Remark: You can also show with definition.

Example — III

$$\begin{aligned} \text{maximize}_{x,y,z} \quad & xyz \\ \text{s.t.} \quad & x + 2y + 3z \leq 3 \\ & x, y, z \geq 0 \end{aligned}$$

- **Answer:** No, because xyz is not a concave function in x, y, z.

Example — III

$$\begin{aligned} \text{maximize}_{x,y,z} \quad & xyz \\ \text{s.t.} \quad & x + 2y + 3z \leq 3 \\ & x, y, z \geq 0 \end{aligned}$$

- **Answer:** No, because xyz is not a concave function in x, y, z.

But we can transform this into maximizing $\log(xyz)$. Then:

$$\begin{aligned} \text{maximize} \quad & \log x + \log y + \log z \leftarrow \text{concave} \\ \text{s.t.} \quad & x + 2y + 3z \leq 3 \qquad \text{(exercise)} \\ & x, y, z \geq 0 \end{aligned}$$

which is a convex optimization problem.

Remarks: Hidden Convexity

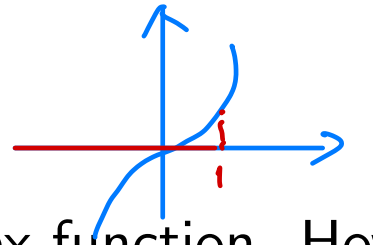
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Examples:

- $\{x : x^3 - 1 \leq 0\}$.

  $g(x) = x^3$ is not a convex function. However, this constraint defines a convex feasible region ($\equiv \{x : x \leq 1\}$).

Remarks: Hidden Convexity

Sometimes the convexity of objective function and constraints are hidden.

Examples:

- $\{x : x^3 - 1 \leq 0\}$.
  $g(x) = x^3$ is not a convex function. However, this constraint defines a convex feasible region ($\equiv \{x : x \leq 1\}$).

- $\{z^2 - xy \leq 0,\ x, y, z \geq 0\}$. $g(x, y, z) = z^2 - xy$ is not a convex function. The Hessian is

$$\nabla^2 g(x, y, z) = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

  with eigenvalues -1, 1 and 2. But this gives a convex region.

Convexity and Optimality

How Does Convexity Help?

Theorem: Convexity and Global Solutions

Let $f : \Omega \to \mathbb{R}$ be a convex function and $\Omega \subseteq \mathbb{R}^n$ be a convex set. Then any local minimizer of the problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{x} \in \Omega \end{aligned}$$

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Proof: By contradiction, assume $\boldsymbol{x}^*$ is a local minimizer, and there exists $\bar{\boldsymbol{x}} \in \Omega$ such that $f(\bar{\boldsymbol{x}}) < f(\boldsymbol{x}^*)$.

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Proof: By contradiction, assume $\boldsymbol{x}^*$ is a local minimizer, and there exists $\bar{\boldsymbol{x}} \in \Omega$ such that $f(\bar{\boldsymbol{x}}) < f(\boldsymbol{x}^*)$. Then, using convexity,

$$f(\lambda \bar{\boldsymbol{x}} + (1-\lambda)\boldsymbol{x}^*) \leq \lambda f(\bar{\boldsymbol{x}}) + (1-\lambda) f(\boldsymbol{x}^*) < f(\boldsymbol{x}^*)$$

for any $0 < \lambda < 1$. This is a contradiction to: $\boldsymbol{x}^*$ is a local min. $\square$

Stationarity and Global Optimality: Unconstrained

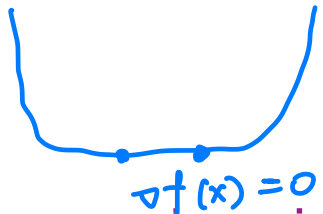


Theorem: Stationarity & Global Optimality

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex on $\mathbb{R}^n$ and continuously differentiable at $\boldsymbol{x}^*$. Then, $\boldsymbol{x}^*$ is a global minimizer of $\min_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x})$ if and only if $\nabla f(\boldsymbol{x}^*) = 0$.

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Remarks:

- In unconstrained convex optimization: stationarity (FONC) is sufficient and necessary for global optimality.

Stationarity and Global Optimality: Unconstrained



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- In unconstrained convex optimization: stationarity (FONC) is sufficient and necessary for global optimality.
- Finding stationary points of a convex optimization problems providing global optimal solutions.

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Remarks:

- In unconstrained convex optimization: stationarity (FONC) is sufficient and necessary for global optimality.
- Finding stationary points of a convex optimization problems providing global optimal solutions.
- In the least squares problem $\min_{\boldsymbol{\beta}} \|\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y}\|^2$, its FONC is $\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top \boldsymbol{y}$. All such $\boldsymbol{\beta}$ are also global minimizers.

Proof: By the first-order characterization of convex function,

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$$

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when $\nabla f(\boldsymbol{x}^*) = 0$, we have

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}^*), \quad \forall \boldsymbol{y} \in \mathbb{R}^n.$$

Proof: By the first-order characterization of convex function,

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when $\nabla f(\boldsymbol{x}^*) = 0$, we have

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}^*), \quad \forall \boldsymbol{y} \in \mathbb{R}^n.$$

The other side is shown by FONC for the unconstrained nonlinear optimization problems.

Theorem: Stationarity & Global Optimality

Consider the constrained convex optimization problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1, \ldots, p, \end{aligned}$$

where f and $\{g_i\}_{i=1}^m$ are convex, and $\{h_j\}_{j=1}^p$ are linear. Then any KKT point for this problem is global minimizers.

Theorem: Stationarity & Global Optimality

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Remarks:

- If f and Ω are convex, then KKT conditions (FONC) are sufficient for global minimizers (no need of CQ at all).

Stationarity and Global Optimality

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- If f and Ω are convex, then KKT conditions (FONC) are sufficient for global minimizers (no need of CQ at all).
- Recall that in order to let the necessary direction (KKT conditions) to be true, we need CQ to hold.

Stationarity and Global Optimality

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where f and $\{g_i\}_{i=1}^m$ are convex, and $\{h_j\}_{j=1}^p$ are linear. Then any KKT point for this problem is global minimizers.

Remarks:

- If f and Ω are convex, then KKT conditions (FONC) are sufficient for global minimizers (no need of CQ at all).
- Recall that in order to let the necessary direction (KKT conditions) to be true, we need CQ to hold.
- If f is concave and Ω is convex, then any KKT points of the problem $\max_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$ are global maximizers of f.

Proof

Let $(\overline{\boldsymbol{x}}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}})$ be a KKT point.

1. Main Condition

$$\nabla f(\overline{\boldsymbol{x}}) + \sum_{i=1}^{m} \overline{\lambda}_i \nabla g_i(\overline{\boldsymbol{x}}) + \sum_{j=1}^{p} \overline{\mu}_j \nabla h_j(\overline{\boldsymbol{x}}) = 0.$$

2. Dual Feasibility

$$\overline{\lambda}_i \geq 0 \quad i = 1, ..., m.$$

3. Complementarity

$$\overline{\lambda}_i \cdot g_i(\overline{\boldsymbol{x}}) = 0 \quad \forall\ i = 1, ..., m.$$

4. Primal Feasibility

$$g_i(\overline{\boldsymbol{x}}) \leq 0, \quad h_j(\overline{\boldsymbol{x}}) = 0 \quad \forall\ i, \quad \forall\ j.$$

Proof

Because f and g_i are convex and h_j is linear, the Lagrangian $\boldsymbol{x} \mapsto L(\boldsymbol{x}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}})$ is convex. The main condition $\nabla_{\boldsymbol{x}} L(\overline{\boldsymbol{x}}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}}) = \mathbf{0}$ implies $\overline{\boldsymbol{x}}$ is a global minimum of $\boldsymbol{x} \mapsto L(\boldsymbol{x}, \overline{\boldsymbol{\lambda}}, \overline{\boldsymbol{\mu}})$ (by the theorem for unconstrained case). Thus,

$$
\begin{aligned}
f(\overline{\boldsymbol{x}}) &= f(\overline{\boldsymbol{x}}) + \sum_{i=1}^{m} \overline{\lambda}_i g_i(\overline{\boldsymbol{x}}) + \sum_{j=1}^{p} \overline{\mu}_j h_j(\overline{\boldsymbol{x}}) \quad \text{(complementarity)} \\
&= \min_{\boldsymbol{x}} \left\{ f(\boldsymbol{x}) + \sum_{i=1}^{m} \overline{\lambda}_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \overline{\mu}_j h_j(\boldsymbol{x}) \right\} \\
&\leq \min_{\{g_i(\boldsymbol{x}) \leq 0, h_j(\boldsymbol{x}) = 0, \forall i,j\}} \left\{ f(\boldsymbol{x}) + \sum_{i=1}^{m} \overline{\lambda}_i g_i(\boldsymbol{x}) + \sum_{j=1}^{p} \overline{\mu}_j h_j(\boldsymbol{x}) \right\} \\
&\leq \min_{\{g_i(\boldsymbol{x}) \leq 0, h_j(\boldsymbol{x}) = 0, \forall i,j\}} f(\boldsymbol{x}).
\end{aligned}
$$

Example

Consider this problem from Tutorial 8 Exercise 3.

check this is Convex optimization.

$$\begin{aligned} \min_{x_1,x_2} \quad & (x_1-3)^2+(x_2-2)^2 \\ \text{s.t.} \quad & 2x_1+x_2-6\le 0 \\ & x_1+2x_2-6\le 0 \end{aligned}$$

- Prove that $x_1 = 11/5$ and $x_2 = 8/5$ is a local minimizer.

Solution: $L(x,\lambda_1,\lambda_2)=(x_1-3)^2+(x_2-2)^2+\lambda_1(2x_1+x_2-6)+\lambda_2(x_1+2x_2-6)=0$

main condition of KKT: $\nabla_x L(x,\lambda_1,\lambda_2)=\begin{bmatrix}2(x_1-3)+2\lambda_1+\lambda_2\\ 2(x_2-2)+\lambda_1+2\lambda_2\end{bmatrix}=0$

plug in $x_1=11/5$, $x_2=8/5$, we have

$$\begin{cases}2\lambda_1+\lambda_2=8/5\\ \lambda_1+2\lambda_2=4/5\end{cases}\Rightarrow\begin{cases}\lambda_1=\frac{4}{5}\\ \lambda_2=0\end{cases}$$

Then check other conditions in KKT also hold. By the previous theorem this point is global minimizer.

Software to Solve Convex Optimization Problems



`MATLAB` has its own functions: `fmincon` and `fminunc`.

- However, in general they are not very scalable nor fast. (One has to provide the right information + adjust parameter).

We suggest to use CVX. CVX can solve a large range of nonlinear optimization problems.

- CVX can only solve convex optimization problems (that is what it is named for).
- It can only recognize certain classes of convex functions.
- Sometimes, one has to manually convert a problem into a recognizable form before inputting into CVX.

Examples

Example 1:

$$\begin{aligned} \text{minimize} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1 \end{aligned}$$

Example 2:

$$\begin{aligned} \text{minimize} \quad & e^{x_1+x_2} + (x_1 - 0.5x_2)^2 + 2.75x_2^2 \\ \text{s.t.} \quad & x_1 + 2x_2 = 1 \end{aligned}$$

```
% Solve the optimization problem
cvx_begin quiet
variables x1 x2

minimize( exp(x1+x2) + (x1-0.5*x2)^2 + 2.75*x2^2 )
subject to
    x1 + 2*x2 == 1;
cvx_end

fprintf(1,'x1 = %g, x2 = %g, opt-val = %g\n',x1,x2,
    cvx_optval);
```

Point Clouds and Circles

Let $\boldsymbol{y}^1, \boldsymbol{y}^2, \ldots, \boldsymbol{y}^k \in \mathbb{R}^2$ be k different points. We want to find a circle in $\mathbb{R}^2$ with minimum radius that contains all of these points:

$$\begin{aligned} \min_{\boldsymbol{y}\in\mathbb{R}^2, r\in\mathbb{R}} \quad & r \\ \text{subject to} \quad & \|\boldsymbol{y}-\boldsymbol{y}^1\| \le r, \ \ \|\boldsymbol{y}-\boldsymbol{y}^2\| \le r, \ \ \ldots \ \ , \|\boldsymbol{y}-\boldsymbol{y}^k\| \le r, \\ & r \ge 0. \end{aligned}$$

$\|y-y^i\|-r$ is a convex function in (y, r)

- This is a convex optimization problem.

Point Clouds: Code

```
% number of points and data points
k = 100; Y = 10*rand(2,k) - 5;

% Solve the optimization problem
cvx_begin quiet
variables r x(2);
    minimize r
        subject to
            norms(x*ones(1,k)-Y,2,1) <= r*ones(1,k);
            r >= 0;
cvx_end

% Plot the results
hold on; axis equal
for i = 1:k
   plot(Y(1,i), Y(2,i),'r.','MarkerSize',12);
end

alpha = 0:pi/50:2*pi;
plot(x(1),x(2),'b.','MarkerSize',12);
plot(x(1)+r*cos(alpha),x(2)+r*sin(alpha),'b','LineWidth',1.5);

hold off;
```

Today's lecture

- Definition of Convex Optimization
- Stationarity and Global Optimality in Convex Optimization

Next lecture

- Algorithms for Unconstrianed Convex Optimization

Questions?